

Dhanush R Giriyan

Research Assistant

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Skills & Relevant Coursework

Languages **Python (most preferred), C++, C, Java, Matlab**

Tools **PyTorch, Keras, TensorFlow, Onnx, Scikit-Learn, Sci-Py**

Relevant Graduate Courses Algebraic Topology; Abstract Algebra; Differential Geometry; Homology; Reinforcement Learning; Combinatorial Algorithms & Optimization; Randomization & Approximation Algorithms; Quantum Computation; Statistical Learning Theory; Combinatorics; Probabilistic Method; Functional Analysis

Education

2023–* **Ph.D. in Computer Science**, Arizona State University, Tempe, AZ, U.S.A

Advised by Prof. Andréa W. Richa

2020–2023 **Computer Science (B.Sc) and Mathematics (B.Sc)**, Arizona State University, Tempe, AZ, U.S.A, graduated with *Summa Cum Laude*

2017–2019 **Civil Engineering (B.Tech)**, National Institute of Technology, Karnataka, India, transferred out

Ongoing Research

Aug 2023–* **Bounding the Mixing Time for the Fixed-Volume Solid-On-Solid model**,
Advised by: Prof. Andréa Richa

- Considered a variant of the model that closely resembles the dynamics at zero temperature
- Defined a potential function that is provably non-increasing as a function of time
- Bounded the expected number of steps before reaching the optimal configuration
- Exploring random walks on high-dimensional expanders

Aug 2024–* **Graph Rewiring with Markov chains for Message Passing Neural Networks**,
Advised by: Prof. Andréa Richa

- Set up a markov chain whose state space is the set of all possible edge sets for a given number of nodes
- Chose a distribution dependent on the total effective resistance of the input graph
- Use the Metropolis-Hastings algorithm to assign transition probabilities
- Currently running experiments

Publications

1. Jamison W. Weber, Dhanush R. Giriyan, Devendra R. Parkar, Dimitri P. Bertsekas, and Andréa W. Richa. Distributed online rollout for multivehicle routing in unmapped environments. In *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems, AAMAS*

- '24, page 1910–1918, Richland, SC, 2024. International Foundation for Autonomous Agents and Multiagent Systems
2. Jessi Cisewski-Kehe, Brittany Terese Fasy, Dhanush Giriyan, and Eli Quist. The weighted euler characteristic transform for image shape classification, 2023

Talks

- Jun 2023 **Image Shape Classification with the Weighted Euler Characteristic Transform (Talk)**, *Symposium on Computational Geometry (2023)*, University of Texas, Dallas, U.S.A
- Jan 2023 **PS2GS: Predicting the 21cm Global Signal directly from Power Spectrum using CNNs (Poster)**, *American Astronomical Society 241 Winter Meeting (2023)*, Virtual
- Oct 2022 **PS2GS: Predicting the 21cm Global Signal directly from Power Spectrum using CNNs (Talk)**, *5th Global 21-cm Workshop (2022)*, University of California, Berkeley, U.S.A
- Apr 2022 **Decentralized Multiagent Rollout (Poster)**, *Fulton Undergraduate Research Initiative Symposium (Spring 2022)*, Arizona State University, U.S.A

Experience

- Spring 2023 **Undergraduate Teaching Assistant**, *Design & Analysis of Algorithms (honors)*, Taught by: *Dr. Joshua Daymude*, CSE 450
- Summer 2022 **Summer REU Student**, *Statistical Approaches to Topological Data Analysis*, Advised by: *Dr. Brittany Fasy*, Montana State University
- Fall 2021 **Grader**, *Design & Analysis of Algorithms*, Taught by: *Prof. Andr  a Richa*, CSE 450

Achievements

- Fall 2023 - * **Fulton Fellow**, *School of Computing & Augmented Intelligence, ASU*
- Spring 2020 - **Dean's List**
- Fall 2022 Awarded for obtaining a GPA above 3.5 each semester with ≥ 12 hours of course work
- Spring 2022 **Fulton Undergraduate Research Initiative (FURI) Scholar**
○ Mentored by *Prof. Andr  a Richa*
- Spring 2017 **Top ~1.4% of the country in JEE-Mains**